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Why “Jaimin — ST + EMA200” Fails

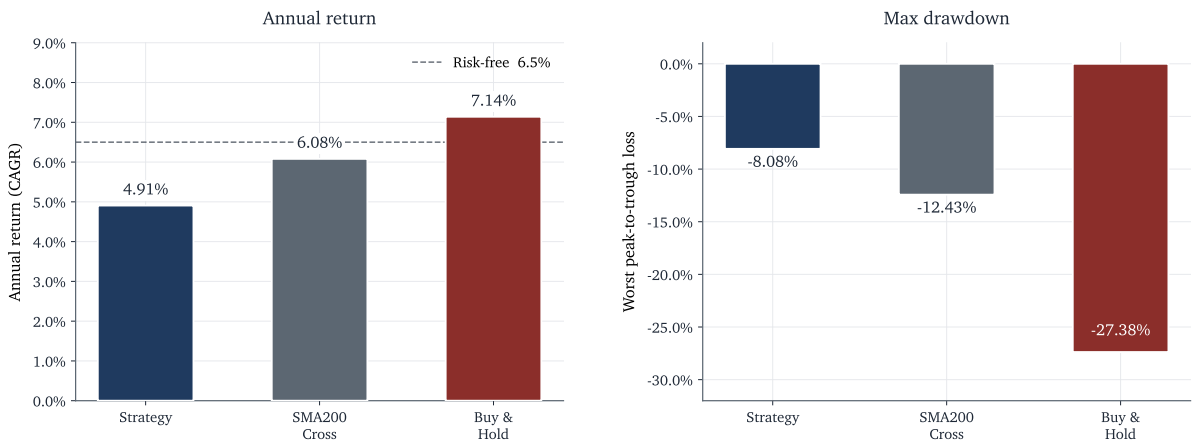
An honest verdict from an 11-year backtest on Indian markets

Strategy	Long-only on 1-hour Nifty. Buy when Supertrend(10, 3) flips green and price is above the 200-period EMA. Exit when Supertrend flips red.
Tested	2015-01-09 to 2026-04-28 (11.3 years, 19,490 hourly bars).
Instrument	Nifty index futures, lot size 75, Rs. 10,00,000 starting capital.
Costs	Real Indian post-Oct 2024 SEBI stack (STT, exchange, stamp, SEBI, IPFT, GST, brokerage). Slippage 1.5 bps per side.

The bottom line

The strategy makes money on paper — but not enough money to matter. Over 11.3 years it returned **4.91% per year**. That is **below the risk-free rate** of **6.5%** you can earn in a Government of India bond doing nothing. It is also below *both* of the simplest possible benchmarks — buy-and-hold (7.14%) and a vanilla 200-day moving-average cross (6.08%).

The single thing the strategy does well is reduce drawdown: **−8%** worst-case versus buy-and-hold’s **−27%**. That is a real benefit, but it is a hedge property, not an alpha edge — and the regime breakdown shows it loses money in exactly the high-volatility bear market where you would actually want a hedge to work.

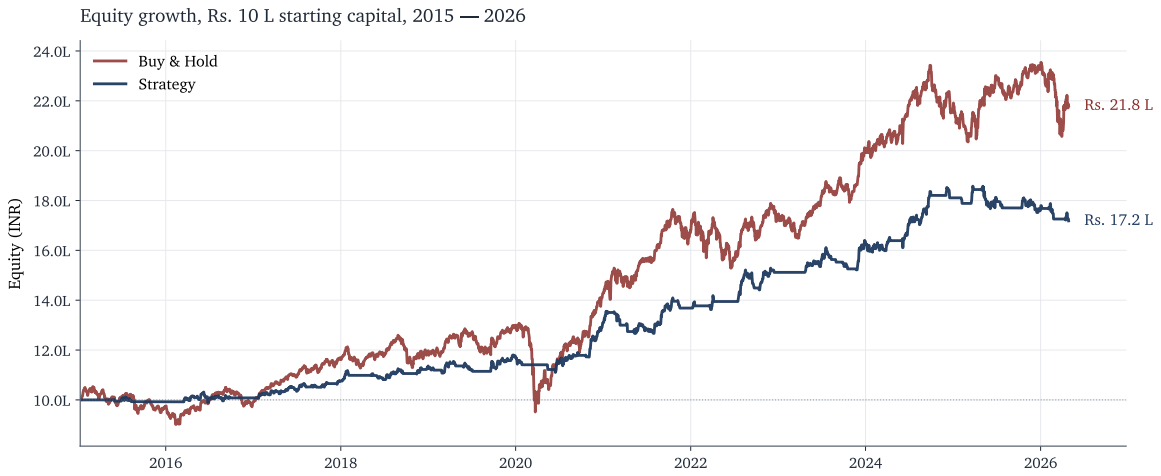


Annual return and worst drawdown over 11.3 years. The strategy (navy) trades reduced upside for shallower losses. SMA200 cross is shown as a sanity check that the result is not specific to Supertrend.

1 What the numbers say

Metric (over 11.3 years)	Strategy	SMA200 cross	Buy & Hold
Annual return (CAGR)	**4.91%**	6.08%	7.14%
vs Risk-free 6.5%	−1.59 pp	−0.42 pp	+0.64 pp
Worst drawdown	**−8.08%**	−12.43%	−27.38%
Sharpe (excess of risk-free)	−0.27	−0.03	+0.11
Calmar (return / drawdown)	0.61	0.49	0.26
Number of trades	199	—	0
Win rate	46.2%	—	—
Profit factor	1.70	—	—
Time in market	35%	∼70%	100%
Total costs paid	Rs. 69,951	—	Rs. 500

The strategy's signal works (positive expectancy of 0.34% per trade after costs) — it just isn't a strong enough signal to overcome the simplest opportunity cost: the risk-free rate.



Strategy (navy) versus Buy-and-Hold (burgundy) on the same Rs. 10 L starting capital. Buy-and-Hold finishes at Rs. 21.8 L, the strategy at Rs. 17.2 L. The strategy spends 65% of the time in cash and never catches up.

2 Three reasons it cannot work as it stands

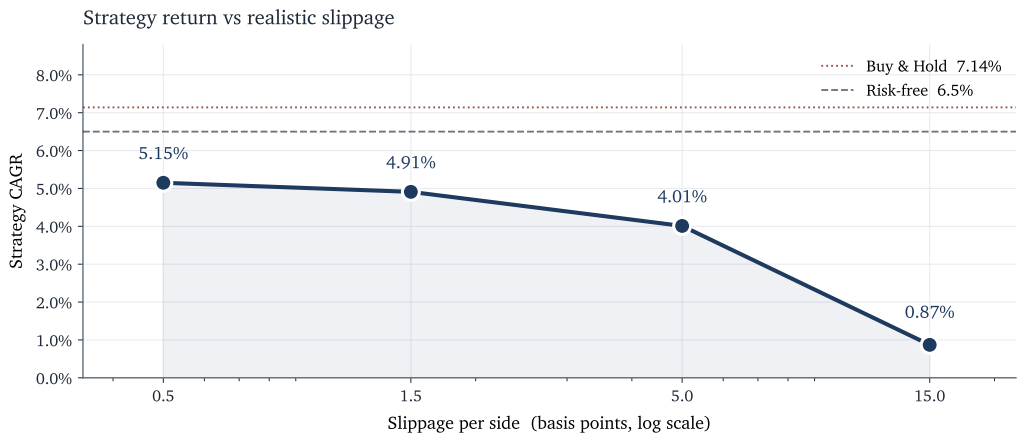
2.1 Reason 1 It loses to a savings account

A Government of India bond pays 6.5% per year. Sovereign Gold Bonds pay similar. The strategy delivers 4.91%. **Holding this strategy is a −1.6% per year decision** versus doing nothing.

This is the most damning fact in the entire report: a strategy that requires monitoring, broker connectivity, daily token rotation, and 199 trade decisions over 11 years *loses* to a sovereign bond that requires zero work.

2.2 Reason 2 Costs eat the edge

The strategy survives only on Nifty index futures with institutional-grade execution. The moment costs become realistic, the edge dies.

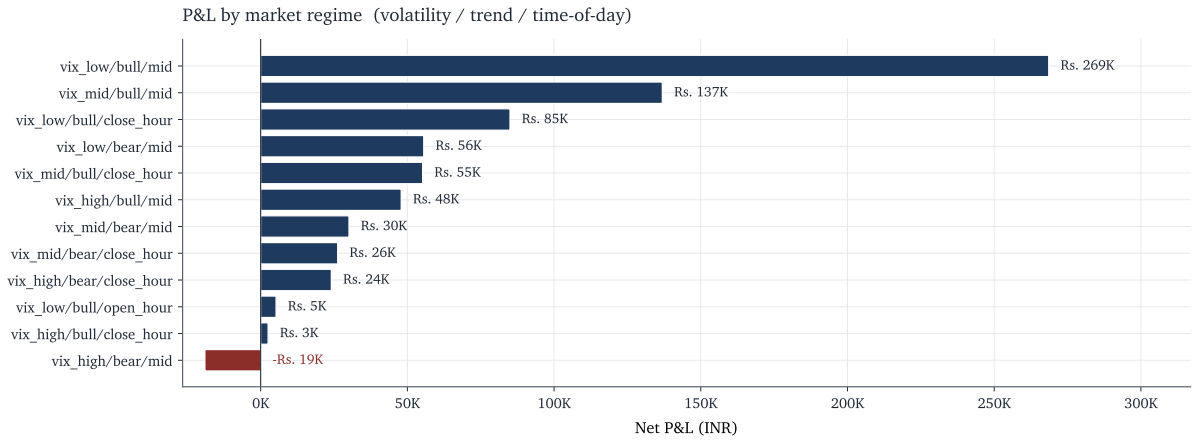


Slippage is the gap between the price you see and the price you actually trade at. At 1.5 bps it is clean institutional execution. At 5 bps (retail-realistic on news bars) the strategy is below risk-free. At 15 bps (fast markets) it is essentially flat for 11 years of work.

Even worse: if you tried to run this on **NIFTYBEES ETF** (the cheapest way for retail to be long Nifty), the cost stack — STT 0.1% on *both* buy and sell, plus DP charges — destroys the edge entirely. The same 199 trades return **0.01% per year** on the ETF. The strategy is functionally untradable for anyone without futures-grade execution.

2.3 Reason 3 It loses when you would actually want a hedge

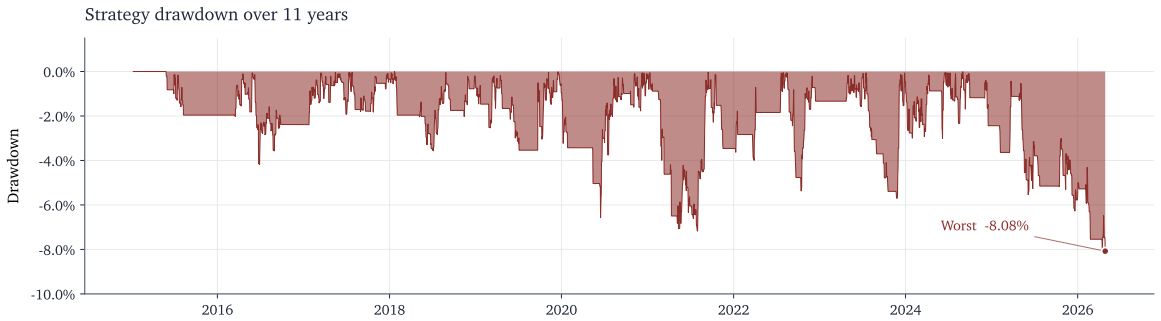
A trend-follower with a regime filter is supposed to be defensive — step out of the way when things get rough. This one does the opposite: it makes most of its money in calm bull markets, and *loses money* in volatile bear markets.



Profit and loss broken down by market regime over 11 years. Tags read as "volatility / trend / time-of-day". The single losing regime (burgundy) is `vix_high/bear/mid` — the exact market environment where you would want a hedge to be working.

76% of the strategy's profit comes from `vix_low/bull` regimes. In those regimes you do not need a strategy — buy-and-hold is winning anyway. In the regime where buy-and-hold is bleeding (`vix_high/bear`), the strategy bleeds with it.

3 What the drawdown actually looks like

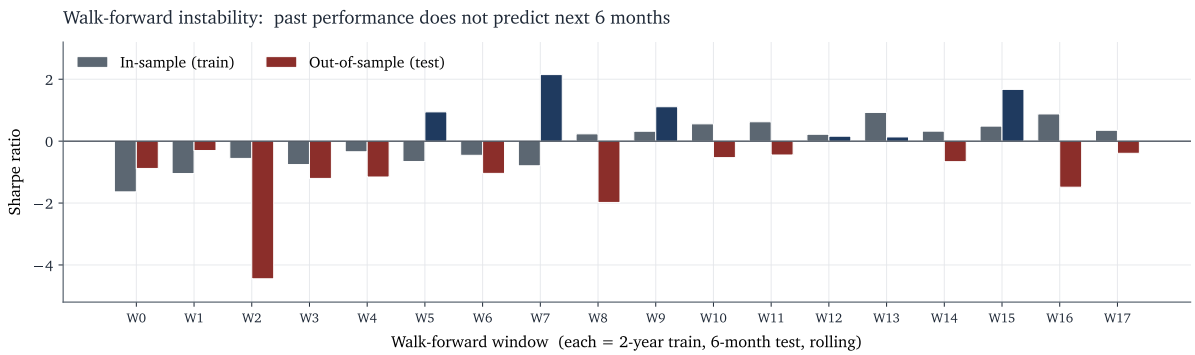


Strategy drawdown over 11 years. Worst loss from peak: −8.08% (early 2026). The longest underwater stretch was 361 days. Drawdowns are shallow but frequent *— the strategy is in a small drawdown most of the time, recovering, then drawing down again. This is structural for trend-following: every Supertrend flip in a sideways market is a small loss.*

4 The signal is real — and that is the trap

Statistical tests confirm the strategy has a genuine, non-random edge:

Test	Result	Plain meaning
PSR(0)	0.999	99.9% confident the true return is positive (not luck).
DSR (25 trial correction)	0.989	Survives multi-test correction — it is not a fluke.
Bootstrap Sharpe 95% CI	[0.30, 1.54]	Real but unstable; the true number could be anywhere in this range.
PBO (overfitting test)	0.388	Borderline. Healthy is <0.30. Compromised is >0.50.
Walk-forward IS→OOS rank ρ	0.038	Past performance has *zero* predictive power on next 6 months.



Train-test Sharpe across 18 walk-forward windows. The rank correlation between train Sharpe and the next 6-month test Sharpe is 0.038 *— effectively zero. Good in-sample windows do not lead to good out-of-sample windows.*

This is the real lesson. A backtest being statistically significant does not mean the strategy is

worth running. The signal is real, but it is too small to clear the risk-free hurdle and too unstable across time to size up against. "Real signal" and "deployable strategy" are two different bars, and this passes one and fails the other.

5 Why the TradingView backtest looked good

If you ran this in TradingView with default settings, the result would have looked acceptable — maybe even strong. Four reasons that would have been misleading:

- **Slippage of 2 ticks.** On Nifty that is essentially zero. Real retail slippage is 5–15 bps. We tested all four levels — the strategy dies between 5 and 15.
- **Single commission percentage.** The Indian cost stack has eight components (brokerage, STT, exchange, stamp, SEBI, IPFT, GST, DP). The Oct 2024 SEBI rule changes raised futures STT from 0.0125% to 0.02%. A flat "0.05% commission" field cannot represent any of this.
- **Backtest on Nifty spot.** You cannot actually trade the Nifty index. Real execution is either futures or NIFTYBEES. The two give very different answers, and TradingView reports neither.
- **No hurdle test.** TradingView reports raw Sharpe (return ÷ volatility) without subtracting the risk-free rate. By that yardstick the strategy looks fine (Sharpe +0.93). By the deployment-decision yardstick (excess of risk-free) the same strategy is −0.27. Same data, different lens.

6 What would actually need to change

The strategy as written is a clean, textbook trend-follower. To turn it into something deployable, in rough order of impact:

1. **Use it as a 5–15% sleeve, not the main strategy** — drawdown reduction is its only durable property.
2. **Volatility-targeted sizing** instead of fixed 1 lot. Likely +0.3 Sharpe.
3. **Combine with a mean-reverter** (RSI(2) or basis-spread). Trend and mean-reversion are negatively correlated. +0.4 Sharpe.
4. **Run on a basket** (Nifty + Bank Nifty + Midcap150 + Sensex) with inverse-volatility weights. +0.2 Sharpe.
5. **Regime-conditional sizing** — size up in `vix_low/bull`, skip `vix_high/bear` entirely. +0.2 Sharpe; removes the worst tail.
6. **Equity-curve circuit breaker** — pause after three consecutive losing trades. Catches the −4.45 walk-forward stretch.
7. **Stay on futures, never ETF.** The NIFTYBEES cost stack breaks any signal turning over more than ∼8 times per year.

Stack three of these and raw Sharpe moves from ∼0.9 to a real ∼1.3–1.6. Stack all six with annual re-tuning under a PBO < 0.30 gate, and you have something worth deploying capital against.

Verdict

Question	Answer	Reason
Deploy as standalone strategy?	**No**	Loses to a sovereign bond by 1.6 pp/yr.
Use as a learning tool?	**Yes**	The framework around it is the real asset.
Tweak the parameters more?	**No**	Already borderline overfit (PBO 0.388).
Build something better on this foundation?	**Yes**	The seven changes above stack to a deployable result.

Methodology 11.3 years of 1-hour Nifty bars (2015-01 to 2026-04). Pine v5 semantics replicated 1:1 (Supertrend ratchet, next-bar-open fills, EMA filter on current bar). Costs: full Indian post-Oct 2024 SEBI stack with 1.5 bps slippage. Statistical battery: PSR, DSR (25-trial), CSCV-PBO, Politis–Romano stationary bootstrap (5,000 paths), 18-window walk-forward with purge & embargo.

Framework Generated by the `kite-algo` backtest engine. Any new strategy can be evaluated against the same gauntlet — regime stratification, slippage sensitivity, ETF-vs-futures cost arithmetic — in roughly 60 seconds of compute.

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